

Time-series models

① Autoregressive model (AR)

The basic idea of an AR model is that the current value of the time-series depends on its past values

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + \varepsilon_t$$

Y_t = current observation

$Y_{t-1}, Y_{t-2}, \dots$ past observations

$\phi_1, \phi_2, \dots, \phi_p \rightarrow$ autoregressive coefficients

c = constant

ε_t = random error at time t

AR(1) model

$$Y_t = c + \phi Y_{t-1} + \varepsilon_t$$

Moving Average Model

→ key idea is that the current value depends on past random shocks or errors.

$$Y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \dots + \theta_q \varepsilon_{t-q}$$

Y_t = current observation

μ = mean of the series

ε_t = current random shock

$\varepsilon_{t-1}, \varepsilon_{t-2}, \dots, \varepsilon_{t-q}$ = past random shocks

$\theta_1, \theta_2, \dots, \theta_q \rightarrow$ coefficients

$$Y_t \leftarrow Y_{t-1} \leftarrow Y_{t-2} \dots \quad [AR \text{ case}]$$

$$\begin{array}{c} \varepsilon_1 \quad \varepsilon_2 \quad \varepsilon_3 \quad \varepsilon_4 \\ \swarrow \quad \uparrow \quad \swarrow \quad \uparrow \quad \swarrow \quad \uparrow \\ Y_2 \quad Y_3 \quad Y_4 \end{array} \quad [MA \text{ case}]$$

MA(1)

Main difference: in AR Y_t is correlated with all previous values

in MA Y_t is correlated with only a few values

AR : depends on past observations
 MA : depends on past errors or shocks.

Intuition behind MA

$$Y_t = \mu + \varepsilon_t + \theta \varepsilon_{t-1}$$

Suppose yesterday had a positive shock $\varepsilon_{t-1} > 0$. Then today's value is affected by that shock through ε_{t-1} .

The effect does not persist indefinitely.

In an $MA(1)$ model, the shock ε_{t-1} affects Y_{t-1} and Y_t but not Y_{t+1} and beyond

An $MA(q)$ model says that a random shock has an effect for at most q periods.

ARMA model

- combines these two ideas

ARMA(p, q) model

$$y_t = c + \phi_1 y_{t-1} + \dots + \phi_p y_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q}$$

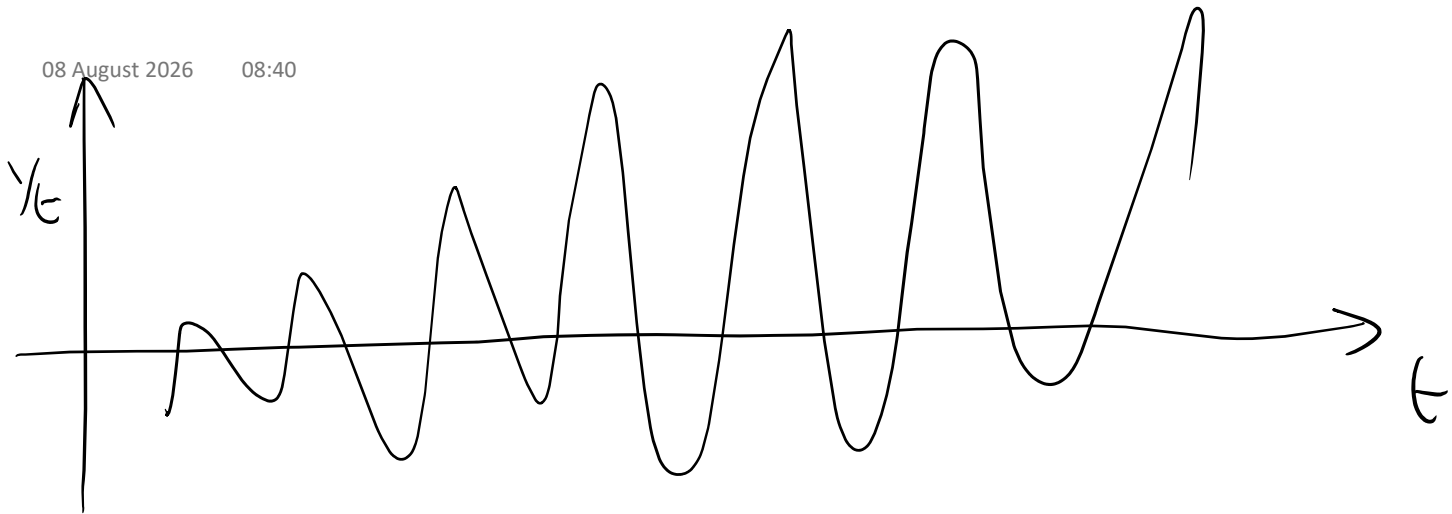
The AR part captures persistence in the series while the MA part captures past shocks.

ARMA(1, 1)

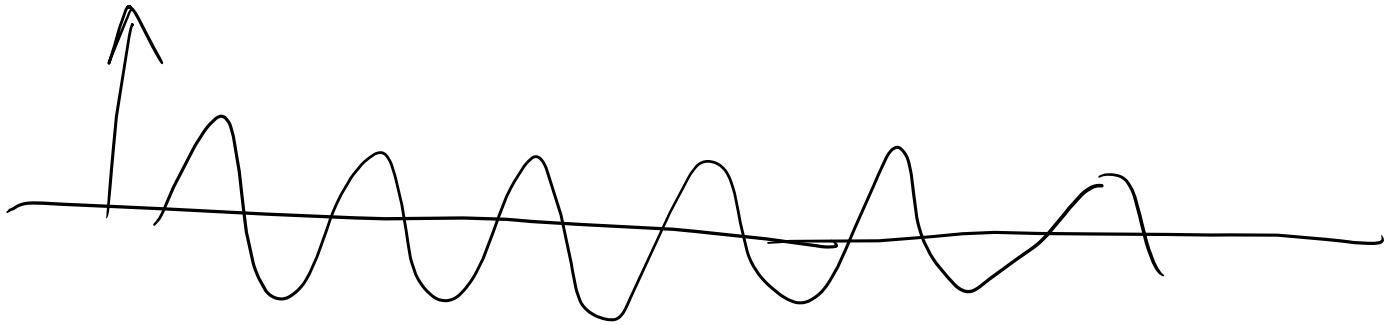
$$Y_t = c + \phi_1 Y_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}$$

When do we use ARMA?

- ARMA models are used
when the time-series is stationary



non-stationary



A stationary time series has statistical properties that do not change with time

→ Mean is stable, variance is stable
autocorrelation structure is stable

100, 105, 110, 115, 123 → strong trend
likely non-stationary.

ARIMA model

Autoregressive Integrated Moving
Average model.

ARIMA(p, d, q) model

p = order of the AR component

q = order of the MA component

d = number of times we difference
the series

ARIMA first makes a non-stationary series more stationary through differencing and then models the resulting series through ARMA

Let the original series be Y_t , then

$$\Delta Y_t = Y_t - Y_{t-1}$$

$$Y_t = 100, 105, 111, 118, 126 \}$$

$$\Delta Y_t = 5, 6, 7, 8$$

ΔY_t is more stationary than Y_t

If one difference is sufficient, then

$$\underline{d = 1}$$

if we need to difference twice

$$\Delta^2 Y_t = \Delta Y_t - \Delta Y_{t-1}, \text{ then } d = 2$$

Putting ARIMA together
 if we have an $ARIMA(p, q, d)$ model

$$\text{Define } X_t = \Delta Y_t = Y_t - Y_{t-1}$$

[Note: Y_t is the original series]

Then X_t follows an $ARMA(1, 1)$ model

$$X_t = c + \phi X_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}$$

$$(Y_t - Y_{t-1}) = c + \phi (Y_{t-1} - Y_{t-2}) + \varepsilon_t + \theta \varepsilon_{t-1}$$

Original series $\xrightarrow[\text{d times}]{\text{differencing}}$ stationary series
 $\downarrow$ $ARMA(p, q)$ model.

how do we identify which model to use?

$$Y_t = \phi Y_{t-1} + \epsilon_t$$

$Y_t, Y_{t-1}, \dots$ will also be correlated



Ordinary autocorrelation measures $\text{Cov}(Y_t, Y_{t-k})$ includes both direct and indirect relationship

The PACF test

After accounting for Y_{t-1} and Y_{t-2} is there still a relationship between Y_t and Y_{t-3}

lag - 3 partial autocorrelation function

lag 1 ACF is the same as ACF

$$PACF(1) = ACF(1)$$

$$PACF(2) \neq ACF(2)$$

Computing the lag-2 PACF

fit $Y_t = \phi_{21} Y_{t-1} + \phi_{22} Y_{t-2} + \varepsilon_t$

↑
regression least-squares

obtain ϕ_{21} and ϕ_{22} through
regression

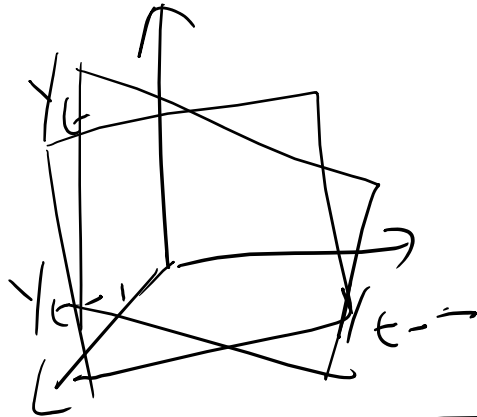
The PACF at lag 2 is ϕ_{22}

The coefficient of the lagged lag in this model ϕ_{22} measures whether

Y_{t-2} has any additional predictive power once Y_{t-1} is already known.

Computing the lag-3 PACF

$$\text{Fit } Y_t = \phi_{31} Y_{t-1} + \phi_{32} Y_{t-2} + \phi_{33} Y_{t-3} + \varepsilon_t$$



$$PACF(3) = \phi_{33}$$

Why does this remove indirect effect?

Suppose the true model is $Y_t = 0.8Y_{t-1} + \varepsilon_t$

Then $Y_{t-1} = 0.8Y_{t-2} + \varepsilon_{t-1}$

$S = Y_{t-2}$ is correlated with Y_t
but only because it influences Y_{t-1}

$$\underline{\text{Corr}(Y_t, Y_{t-2}) > 0}$$

Now fit

$$\underline{Y_t} = \underline{aY_{t-1}} + \underline{bY_{t-2}} + \underline{\varepsilon_t}$$

Since Y_{t-1} already captures all the predictive information from Y_{t-2} , the regression will

find $b \approx 0$ $\therefore$ $PACF(2) \approx 0$

[regression will find $a \approx 0.8, b \approx 0$]

because the actual model is

$$Y_t = 0.8Y_{t-1} + \varepsilon_t$$

In an AR(1) model, ρACF will cut off
after lag 1.

$$\rho ACF(1) = ACF(1)$$

$$\rho ACF(2) = 0$$

$$\rho ACF(3) = 0$$

A numerical example

t Time	Y_t Value
1	3
2	5
3	4
4	6
5	7
6	8

To calculate the lag-2 PACF we would fit

$$Y_t = aY_{t-1} + bY_{t-2} + \varepsilon_t$$

to the data on the left

Y_t	Y_{t-1}	Y_{t-2}
$Y_3 = 4$	$Y_2 = 5$	$Y_1 = 3$
$Y_4 = 5$	$Y_3 = 4$	$Y_2 = 5$
$Y_5 = 7$	$Y_4 = 6$	$Y_3 = 4$
$Y_6 = 8$	$Y_5 = 7$	$Y_4 = 6$

$$\frac{\partial E}{\partial \phi_1} = 0$$

$$\frac{\partial E}{\partial \phi_2} = 0$$

Now fit a regression curve

to this data

Squared error

$$E = f(\phi_1, \phi_2)$$

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2}$$

For a MA model, ACF does not cut off abruptly

In a MA model, ACF cuts off abruptly after some lags but $PACF$ does not cut off

In an AR model, ACF does not cut off, but $PACF$ cuts off

$\rho_{A \bar{F}}$ (finally)

$$\rho_{A \bar{F}}(3) = \frac{\text{Cov}(X_t, X_{t-3} \mid \overset{\text{given}}{X_{t-1}, X_{t-2}})}{\sqrt{\text{Var}(X_t \mid X_{t-1}, X_{t-2})} \sqrt{\text{Var}(X_{t-3} \mid X_{t-1}, X_{t-2})}}$$

Let us try to compute $1ACF(z)$ through
representation.

$$X_t = a_1 X_{t-1} + a_2 X_{t-2} + e_t$$

$$X_{t-3} = b_1 X_{t-1} + b_2 X_{t-2} + f_t$$

residual here is $e_t = X_t - \hat{X}_t$

residual here is $f_t = X_{t-3} - \hat{X}_{t-3}$

e_t is the part of X_t that cannot be predicted from X_{t-1} and X_{t-2}

f_t is the part of X_{t-3} that cannot be explained by X_{t-1} and X_{t-2}

$$\text{Cov}(e_t, f_t) = \text{Cov}(X_t, X_{t-3} | X_{t-1}, X_{t-2})$$

$$\text{Var}(X_t | X_{t-1}, X_{t-2}) = \text{Var}(e_t)$$

$$\text{Var}(X_{t-3} | X_{t-1}, X_{t-2}) = \text{Var}(f_t)$$

$$\rho_{ACF}(3) = \text{Cov}(e_t, f_t)$$

$$\text{Cov}(e_t, f_t) = \frac{\text{Cov}(e_t, f_t)}{\sqrt{\text{Var}(e_t)} \sqrt{\text{Var}(f_t)}} ;$$

$$-1 \leq \text{Cov}(e_t, f_t) \leq 1$$

$$\rho_{ACF}(3) = \text{Cov}(e_t, f_t)$$

$PACF(3) = \text{Cov}(e_t, f_t)$ is equal to
the coefficient of X_{t-3} in
the regression

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \boxed{\phi_3} X_{t-3} + \varepsilon_t$$

$$\boxed{PACF(3) = \text{Cov}(e_t, f_t) = \phi_3}$$

Dickey-Fuller test

↳ we want to check if a time-series is stationary or not

Consider an AR(1) process:

$$Y_t = \rho Y_{t-1} + \varepsilon_t$$

Case 1: $|\rho| < 1$

The process is stationary

$$Y_t = 0.8 Y_{t-1} + \varepsilon_t$$

A shock to the series gradually dies out in time

$$Y_{t+1} = 0.8 Y_t + \varepsilon_{t+1}$$

$$= 0.8(0.8 Y_{t-1} + \varepsilon_t) + \varepsilon_{t+1}$$

$$Y_{t+1} = (0.8)^2 Y_{t-1} + 0.8 \varepsilon_t + \varepsilon_{t+1}$$

Case 2: $\rho = 1$

$$Y_t = Y_{t-1} + \varepsilon_t, \quad \Delta Y_t = Y_t - Y_{t-1} = \varepsilon_t$$

→ a random walk

$$Y_t = \varepsilon_1 + \varepsilon_2 + \dots + \varepsilon_t$$

Var(Y_t) keeps growing → non-stationary

Case 3: $|\rho| > 1$

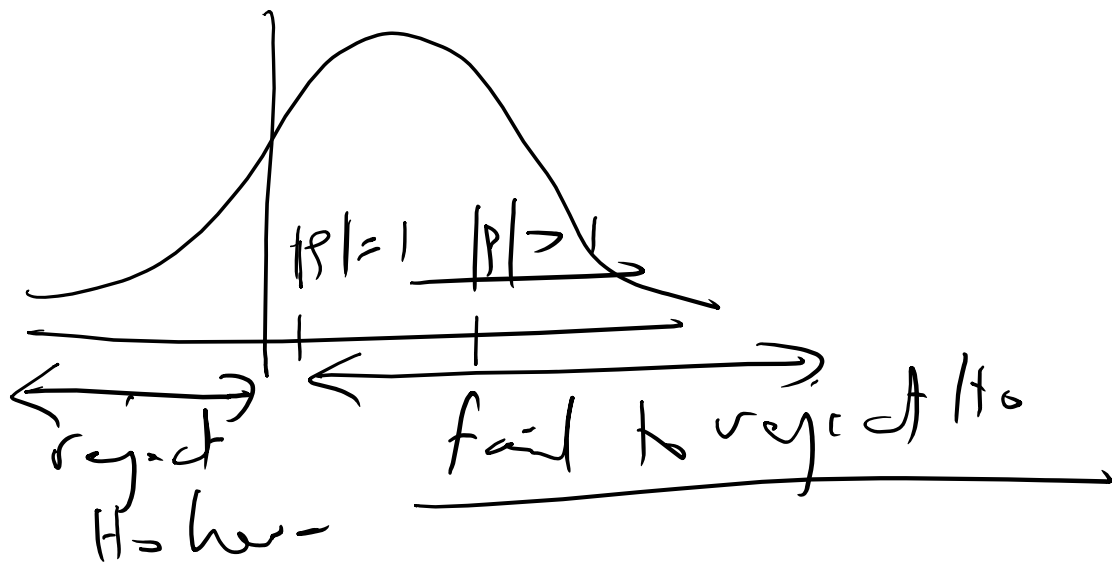
the process is explosive & non-stationary

The null and alternative hypotheses

$$\underbrace{H_0: |\rho| = 1}_{\text{non-stationary}} \quad / \quad \underbrace{H_1: |\rho| < 1}_{\text{stationary}}$$

$|\rho| > 1$ is covered in H_0

Dickey Fuller is a left-tailed test



We are talking about unitary roots
of some polynomial?

Let us build this polynomial

$$Y_t = \phi Y_{t-1} + \varepsilon_t$$

Introduce a backshift operator B

$$B Y_t = Y_{t-1}, \quad B^2 Y_t = Y_{t-2}$$

$$Y_t = \phi B Y_t + \varepsilon_t$$

$$(1 - \phi B) Y_t = \varepsilon_t$$

$(1 - \phi B)$ is called the AR polynomial

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \varepsilon_t$$

$$\phi(B) = \frac{1 - \phi_1 B - \phi_2 B^2}{1}$$

for $AR(p)$

we have $\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p$

$$\phi(B) Y_t = e_t$$

$$(1 - \phi B) Y_t = e_t \text{ in case of } AR(1)$$

Can we write $Y_t = \boxed{(1 - \phi B)^{-1}} e_t$

What do we mean by $(1 - \phi B)^{-1}$?

We mean an operator T such that $(1 - \phi B)T = I$

$(1 - \phi B)^{-1}$ is an inverse operator

Can we guess this series?

$$(1 - \phi B)^{-1} = 1 + \phi B + \phi^2 B^2 + \dots$$

$$\begin{aligned} & \cancel{(1 - \phi B)} (1 + \phi B + \phi^2 B^2 + \dots) = \frac{1}{} \\ & \underline{1 - \cancel{\phi B} + \cancel{\phi B} - \cancel{\phi^2 B^2} + \cancel{\phi^2 B^2} - \cancel{\phi^3 B^3} + \cancel{\phi^3 B^3}} \end{aligned}$$

$$(1 - \phi B) Y_t = \varepsilon_t$$

$$Y_t = (1 - \phi B)^{-1} \varepsilon_t$$

$$Y_t = (1 + \phi B + \phi^2 B^2 + \dots) \varepsilon_t$$

$$Y_t = \boxed{\varepsilon_t} + \phi \varepsilon_{t-1} + \phi^2 \varepsilon_{t-2} + \dots$$

$\text{Var}(Y_t)$ is finite for all $t \rightarrow$ when?

$$\text{Var}(Y_t) = \sigma^2 \sum_{j=0}^{\infty} \phi^{2j} \quad \underline{\text{as } t \rightarrow \infty.}$$

Since $\varepsilon_t, \varepsilon_{t-1}, \dots$ are all

independent
shocks.

unless $\sum_{j=0}^{\infty} \phi^{2j}$ is a constant, the variance
will explode to ∞

when $\boxed{|\phi| < 1} \rightarrow \sum_{j=0}^{\infty} \phi^{2j} = 1 + \phi^2 + \phi^4 + \dots$
 $= \frac{1}{1 - \phi^2}$

provided $|\phi| < 1$

$1 - \phi B$ is the autoregressive polynomial
for AR(1)

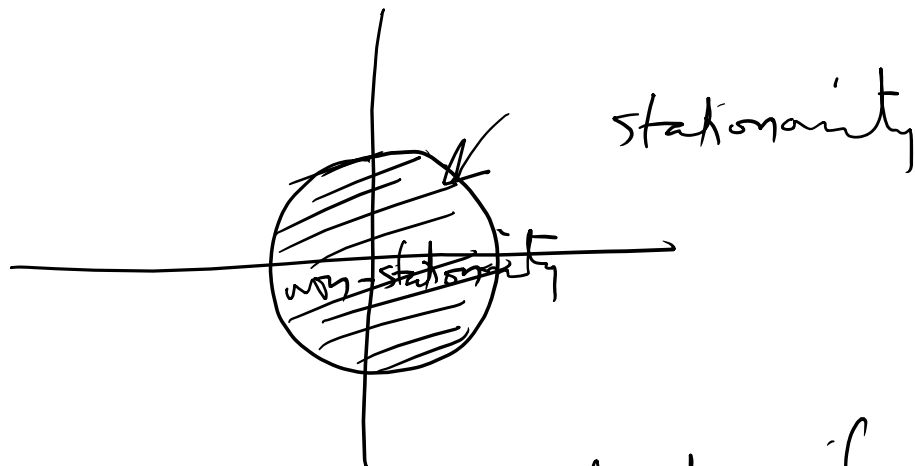
Suppose we want to find the roots
of this polynomial

$$1 - \phi B = 0 \Rightarrow B = \frac{1}{\phi}$$

For stationarity $|\phi| < 1 \Rightarrow \underline{B > 1}$

for non-stationarity $|\phi| > 1 \Rightarrow B < 1$

Whenever the roots of the AR
polynomial lie outside the unit
circle, we have stationarity
otherwise we have non-stationarity



The DF test check, if the root lies on the unit circle

if the root lies inside the unit circle, the DF test will still conclude non-stationarity